

Moment Determinacy and Nonlinear Transseries

A sharp Gevrey threshold, exact flat defects,
convergent ambiguity sectors, and Lambert– W folds

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Abstract

A complete formal expansion need not identify the function that is to be inverted, even when the function is a Stieltjes transform of a positive probability measure. We study the nonlinear realization problem $T(y) = yS(T(y))$, where $S(t) = \int_0^\infty (1 + t\lambda)^{-1} d\mu(\lambda)$. First, inverse identifiability is shown to be exactly equivalent to Stieltjes moment determinacy. A cancellation-free Lagrange formula preserves the exact factorial growth order of the moment sequence. Consequently, formal Gevrey order at most two forces uniqueness in this positive-measure class, whereas every order greater than two admits explicit continua of distinct realizations.

Two complementary families make the hidden data explicit. Generalized gamma measures give an exact defect $Ct^{-\beta} \exp(-At^{-\alpha})$; bilateral geometric measures give a reciprocal theta function with a nonconstant log-periodic amplitude. For the latter, the ambiguity is comparable to the least moment-remainder bound. We transport both defects through nonlinear inversion, calculate the first relative correction, and construct convergent expansions in successive powers of the flat defect with explicit Cauchy bounds. After a defect-dependent rescaling, the deformation map converges to ve^{-v} . This proves an asymptotically exact radius for the parameter expansion, Cayley-number limits for its coefficients, and a nearby square-root fold, including its first gamma and log-periodic corrections.

The moment-indeterminacy constructions, Hardy-type determinacy argument, Jacobi identity, and Lagrange inversion are classical foundations. The paper develops their quantitative nonlinear synthesis, without claiming exhaustive historical priority or a solution of the repository’s general Hahn-series realization problem. All proofs are conventional mathematical proofs, not Lean formalizations. Reproducible finite checks and a precise research agenda accompany the article.

Scope. Positive-real asymptotic germs as $y \downarrow 0$; formal coefficient growth; explicit analytic realizations and inverse parameter singularities. No general summation rule is assumed.

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1 The realization question and the contribution

1.1 A focused question arising from the repository

The ProveIt transseries corpus distinguishes formal algebra, asymptotic scales, flat functions, and analytic inversion certificates [1, 2]. These distinctions suggest a precise realization question:

If every coefficient of an inverse expansion is known, and the underlying function is required to come from a positive measure, is the actual inverse uniquely determined?
If not, what is the size and analytic structure of the missing information?

This question is narrower than the construction of a general field of transseries and broader than computing another list of inverse coefficients. It admits a complete answer at the level of identifiability, a sharp coefficient-growth threshold, and explicit models of the remaining ambiguity.

The inspected repository companion on action accumulation already exhibits flat oscillations and studies their transport under inversion [3]. Its editorial notes explicitly distinguish those results from the canonical volume’s general realization question, identified there by the label `plt:rmk:ext-open-realization`. We maintain that distinction. Here the support is that of a *positive Stieltjes moment problem*; we do not construct a realization of every left-finite Hahn series. The positive measure is the additional structure, and moment determinacy is exactly the extra condition that restores identifiability.

Editorial note (ProveIt, 2026-09-29). Two filed repository results lie on the determinate side of Theorem 3.2. The canonical volume [11] (its Theorem `p8:thm:bell`) writes the tail $C(x) = e^{-1} \sum_{j \geq 0} 1/(j!(x+j+1))$ as a Stieltjes transform whose moments are the Bell numbers, with a signed remainder. In the present notation $C(x) = x^{-1} S_\mu(1/x)$ for the probability measure $\mu = \sum_{j \geq 0} \delta_{j+1}/(e j!)$, with moments $m_n = \mathfrak{B}_{n+1} \leq (n+1)!$, of Gevrey order at most one; so Lemma 3.4 makes that representation the only positive one, and by Theorem 3.2 the corresponding inverse has exactly one positive Stieltjes realization. The certified $q \rightarrow 1$ package [12] (its `thm:stieltjes` and `thm:pade`) writes a finite-size correction as $\int d\nu_\tau(u)/(1+su)$ with a positive measure whose moments are bounded by $CA^{2j}(2j)!$, that is, of factorial-square growth, the boundary case $s = 2$ of Theorem 3.5; it proves determinacy by the argument of Lemma 3.4 and adds two-sided Padé bounds. The inverse-digamma package [13] has a Stieltjes-type representation only modulo a convergent Laurent function, outside the class used here. The word *realization* in this article means a positive Stieltjes representation of the inverse germ at $y \downarrow 0$, not the Hardy-field sense of the canonical volume’s Definition `plt:def:cmp-realization`. The limit $v_0 = -W_0(-\kappa)$ of (8.5) is that volume’s Cayley tree function (`p1:def:cayley`). This note is editorial.

1.2 The main results in one place

Let $m_n = \int \lambda^n d\mu$, $m_0 = 1$, and write the formal inverse as

$$\hat{T}(y) = \sum_{n \geq 1} b_n y^n, \quad c_n = (-1)^n b_{n+1}. \quad (1.1)$$

The following statements will be proved in order.

An exact identifiability criterion. The actual germs T_μ and the representing measures μ are in one-to-one correspondence. All measures with the same moments produce the same formal inverse. Thus that formal inverse identifies a unique positive Stieltjes realization if and only if its moment problem is determinate (Theorem 3.2).

A sharp factorial-growth threshold. The coefficients satisfy the noncancelling identity

$$c_n = \frac{1}{n+1} \sum_{k_1+\dots+k_{n+1}=n} m_{k_1} \cdots m_{k_{n+1}}. \quad (1.2)$$

In particular, $m_n \leq c_n$, and a bound $m_n \leq C A^n (n!)^s$ implies $c_n \leq (4AC)^n (n!)^s$. Formal Gevrey order $s \leq 2$ therefore forces uniqueness. For every $s > 2$, a generalized gamma family gives distinct positive realizations of exact formal Gevrey order s (Theorems 3.5 and 4.1).

Explicit missing data. The generalized gamma family has a closed exponential defect, while the bilateral geometric family has a reciprocal theta defect. The latter has a positive log-periodic amplitude and is comparable to the optimally chosen moment-remainder bound (Theorems 4.2, 5.2, and 5.3). Even an exact first-order q -difference equation, added to positivity and the complete formal expansion, need not select a unique transform (Proposition 5.5).

Nonlinear sectors and their radius. For a physical baseline $t_0 = T_{a_0}(y)$, put $D_0 = D(t_0)$ and let $p \rightarrow \infty$ be the logarithmic slope scale of the defect. The expansion of $T_{a_0+\delta}(y)$ in δ converges, with explicit coefficient and remainder bounds. Its radius R_y obeys

$$R_y \sim \frac{t_0}{e p y D_0}. \quad (1.3)$$

At fixed sector number k , the normalized coefficient tends to $k^{k-1}/k!$. The critical deformation is a Lambert– W fold, and its first correction retains the theta phase (Theorems 7.1, 8.2, and 8.5).

1.3 What is classical, and what is claimed here

Moment-indeterminate generalized gamma and Stieltjes–Wigert measures are classical; useful primary accounts include [4, 5, 6, 7]. The order-two determinacy boundary is a Hardy/Carleman phenomenon, not a newly discovered boundary in moment theory. Likewise, Jacobi’s triple product, Lagrange inversion, and the local square-root normal form are classical tools. They are either stated with attribution or derived in the form used below.

The contribution developed in this report is the organized nonlinear realization theorem, its cancellation-free coefficient control, the exact defect transport with relative corrections, the quantitative convergent sector expansion, and the defect-dependent parameter-radius and fold analysis. These are proposed research contributions with proofs; we do not assert that no equivalent formulation has appeared elsewhere. The source comparison was targeted, not a theorem-by-theorem audit of the large repository corpus. No result in this report is claimed to have been checked by Lean.

It is also important not to confuse the present nonuniqueness with a failure of a sectorial implicit-function theorem. A prescribed Borel-summable class contains additional analytic information; for example, Nikolaev’s theorem concerns a selected summable asymptotic setting [9]. Our input data are positivity and a full formal jet, without such a selection rule. The various q -summation procedures also have their own hypotheses [8]; none is silently identified with the families below.

2 Positive Stieltjes transforms and their inverse germs

Definition 2.1. Let $\mathcal{P}_\infty$ be the probability measures on $[0, \infty)$ with finite moments of every nonnegative integer order. For $\mu \in \mathcal{P}_\infty$, define

$$S_\mu(t) = \int_0^\infty \frac{d\mu(\lambda)}{1+t\lambda}, \quad F_\mu(t) = \frac{t}{S_\mu(t)} \quad (t > 0). \quad (2.1)$$

The *positive Stieltjes inverse* is $T_\mu = F_\mu^{-1}$. Thus its defining equation is

$$T_\mu(y) = yS_\mu(T_\mu(y)). \quad (2.2)$$

A flat function at zero is one that is $O(t^N)$ for every N .

Lemma 2.2 (Regularity, remainder, and positivity). *The function S_μ is holomorphic on $\Re t > 0$, is positive on $(0, \infty)$, and extends smoothly from the right to $t = 0$. For every $k \geq 0$,*

$$S_\mu^{(k)}(t) = (-1)^k k! \int_0^\infty \frac{\lambda^k}{(1+t\lambda)^{k+1}} d\mu(\lambda). \quad (2.3)$$

In particular, $0 < S_\mu(t) \leq 1$ and $S'_\mu(t) \leq 0$ for real $t > 0$. For every integer $N \geq 0$,

$$S_\mu(t) = \sum_{n=0}^{N-1} (-1)^n m_n t^n + R_N(t), \quad (2.4)$$

$$R_N(t) = (-t)^N \int_0^\infty \frac{\lambda^N}{1+t\lambda} d\mu(\lambda), \quad 0 \leq (-1)^N R_N(t) \leq m_N t^N. \quad (2.5)$$

The empty sum is zero when $N = 0$.

Proof. On the right half-plane, $|1+t\lambda| \geq 1$. Differentiation on compact sets is justified by the finite moments, giving (2.3); the same domination gives continuity of every right derivative at zero. Holomorphy follows from dominated local holomorphic integration. The signs are immediate. Finally, integrate the finite geometric identity $(1+z)^{-1} = \sum_{n < N} (-z)^n + (-z)^N/(1+z)$ with $z = t\lambda \geq 0$. No infinite geometric expansion has been used. $\square$

Proposition 2.3 (Existence and a residual certificate). *The map F_μ is an increasing bijection of $(0, \infty)$ onto itself. Equation (2.2) has exactly one positive solution for every $y > 0$. It satisfies*

$$0 < T_\mu(y) \leq y, \quad T_\mu(y) \geq y(1 - m_1 y) \quad \text{when } m_1 y \leq 1. \quad (2.6)$$

For any trial value $v > 0$,

$$|v - T_\mu(y)| \leq |v - yS_\mu(v)|. \quad (2.7)$$

The inverse is smooth from the right at zero, with

$$T_\mu(y) = y - m_1 y^2 + (m_1^2 + m_2) y^3 + O(y^4). \quad (2.8)$$

Proof. Since

$$F'_\mu(t) = \frac{S_\mu(t) - tS'_\mu(t)}{S_\mu(t)^2} \geq 1,$$

F_μ is strictly increasing. Moreover, $F_\mu(t) \rightarrow 0$ at zero and $F_\mu(t) \geq t \rightarrow \infty$. The upper root bound follows from $S \leq 1$. The identity $S(t) \geq 1 - m_1 t$, together with $T \leq y$, gives the lower bound. For fixed y , the residual function $H(t) = t - yS_\mu(t)$ has derivative at least one on the positive axis. The real mean-value theorem proves (2.7).

At the boundary point $(t, y) = (0, 0)$, the equation has derivative one in t . Repeated one-sided implicit differentiation gives smoothness and determines every Taylor coefficient. Equivalently, extend each finite smooth jet across the endpoint and apply the finite-order implicit-function theorem. Substitution to degree three yields (2.8). $\square$

Remark 2.4 (What a residual certificate does not know). The estimate (2.7) evaluates the actual transform. It does not turn a formal coefficient list into a selection of μ . The information-theoretic lower bound in Section 6 will show why that distinction cannot be omitted.

3 Identifiability and the sharp Gevrey threshold

Definition 3.1. A formal series $\sum a_n z^n$ has *Gevrey order at most s* if $|a_n| \leq CA^n(n!)^s$ for some finite $C, A > 0$. It has *exact order s* if it has order at most s , but no smaller order. We use this coefficient convention, not a convention shifted by one derivative. An assertion about this formal order alone does not assert a sectorial Borel summation theorem.

Theorem 3.2 (Inverse identifiability equals moment determinacy). *Fix a Stieltjes moment sequence $m = (m_n)_{n \geq 0}$, with $m_0 = 1$, that has at least one representing measure in $\mathcal{P}_\infty$. Every representing measure produces the same formal inverse*

$$\hat{T}(y) = y\hat{S}(\hat{T}(y)), \quad \hat{S}(t) = \sum_{n \geq 0} (-1)^n m_n t^n. \quad (3.1)$$

Different representing measures produce different actual inverse germs. Conversely, the formal inverse uniquely determines m . Hence a unique positive Stieltjes inverse realization exists exactly when the moment problem is determinate.

Proof. Finite Taylor substitution in Lemma 2.2 and Proposition 2.3 gives (3.1) to every finite order. In the formal power-series ring, the equation recursively determines a unique series beginning with y .

Suppose two actual inverse germs agree. Inverting their increasing functions shows that their transforms agree on an interval adjacent to zero. The identity theorem extends equality to the right half-plane. At $t = 1$, equality of all derivatives implies equality of

$$\int \frac{\lambda^k}{(1 + \lambda)^{k+1}} d\mu(\lambda) \quad (k \geq 0).$$

Push μ forward by $u = \lambda/(1 + \lambda)$, and multiply the resulting measure by $1 - u$. We obtain a finite measure on $[0, 1]$ whose polynomial moments are the displayed integrals. Polynomial approximation of continuous functions on $[0, 1]$ gives equality of these weighted measures. Neither original push-forward has an atom at $u = 1$, and the weight is strictly positive on $[0, 1)$. Division by the weight, first on $[0, 1 - \varepsilon]$ and then by monotone exhaustion, recovers equality of the original measures.

Finally, $\hat{T}(y) = y + O(y^2)$ has a unique formal compositional inverse $\hat{F}$. The relation $\hat{S}(t) = t/\hat{F}(t)$ reconstructs every moment. This also shows that no ambiguity is introduced by passing from the transform jet to the inverse jet. $\square$

Lemma 3.3 (A cancellation-free coefficient formula). *Write $\hat{T}(y) = \sum_{j \geq 1} b_j y^j$ and $c_n = (-1)^n b_{n+1}$. Then (1.2) holds, $c_n \geq m_n$, and, for $s \geq 0$, $C \geq 1$,*

$$m_n \leq CA^n(n!)^s \implies c_n \leq (4AC)^n(n!)^s \quad (n \geq 1). \quad (3.2)$$

Thus the moment sequence and the inverse coefficient sequence have exactly the same infimum of admissible Gevrey orders; if the order is attained for either sequence, it is attained for both.

Proof. The formal Lagrange coefficient identity gives

$$b_{n+1} = \frac{1}{n+1} [t^n] \widehat{S}(t)^{n+1}.$$

Every product contributing to degree n has sign $(-1)^n$, so there is no cancellation. The $n+1$ terms with exactly one index equal to n contribute m_n after division by $n+1$. For the upper bound, at most n indices in a weak composition are positive, and $m_0 = 1$. Each product is therefore at most $C^n A^n (\prod_j k_j!)^s \leq C^n A^n (n!)^s$. There are $\binom{2n}{n} \leq 4^n$ weak compositions of n into $n+1$ parts. Division by $n+1$ only improves the bound. $\square$

Lemma 3.4 (The factorial-square determinacy criterion). *A Stieltjes moment sequence satisfying $m_n \leq C A^n (n!)^2$ is determinate.*

Proof. For a representing random variable $X \geq 0$, let U have the symmetric law obtained from $\sqrt{X}$ by choosing either sign with probability one half. Its even moments are m_n , and its odd moments vanish. Choose $c > 0$ with $c^2 A < 1$. Monotone convergence gives

$$\mathbb{E} \cosh(cU) = \sum_{n \geq 0} \frac{c^{2n} m_n}{(2n)!} < \infty,$$

since $(n!)^2 \leq (2n)!$. Thus $\mathbb{E} e^{c|U|} \leq 2\mathbb{E} \cosh(cU) < \infty$. The moment generating function is holomorphic in the strip $|\Re z| < c$, and its Taylor series at zero is fixed by the moments. Any two representing measures therefore yield identical moment generating functions near zero and, by the identity theorem, on the strip. Their characteristic functions agree on the imaginary axis. Uniqueness of the Fourier transform of a finite measure gives equality of the symmetric laws. Squaring recovers the same law of X . The exponential integrability argument applies to every competitor because it uses only the shared moments. $\square$

Theorem 3.5 (Sharp inverse Gevrey threshold). *In the class of positive Stieltjes inverses, a realizable formal inverse of Gevrey order at most $s \leq 2$ has exactly one realization. For every real $s > 2$, there are uncountably many distinct realizations of a single formal inverse of exact Gevrey order s .*

Proof. By Lemma 3.3, a Gevrey bound on the inverse implies the same bound on the moments. For $s \leq 2$, this is a factorial-square bound, so Lemma 3.4 and Theorem 3.2 give uniqueness. For $s > 2$, take $\alpha = 1/s$ in Theorem 4.1 below. That theorem gives a continuum of distinct measures with common moments of exact order $1/\alpha = s$. Lemma 3.3 transfers the exact order, and Theorem 3.2 transfers distinctness. $\square$

Remark 3.6 (The precise sense of sharpness). The threshold is a universal *sufficient* growth condition within a specified positive-measure realization class. It does not say that every sequence of order greater than two is indeterminate. Nor does it say that arbitrary smooth or sectorial analytic functions of order at most two are determined by their positive-real formal expansions. The classical component is Lemma 3.4; compare the treatment of Hardy's condition and generalized gamma laws in [4].

4 Generalized gamma measures and an exact exponential defect

Fix parameters

$$0 < \alpha < \frac{1}{2}, \quad \beta > 0, \quad a = \frac{\beta}{\alpha}, \quad c = \cos(\pi\alpha), \quad A = \sec(\pi\alpha). \quad (4.1)$$

Define

$$f(\lambda) = \frac{\alpha}{\Gamma(a)} \lambda^{\beta-1} e^{-\lambda^\alpha}, \quad (4.2)$$

$$p_*(\lambda) = \sin(\tan(\pi\alpha)\lambda^\alpha - \pi\beta), \quad (4.3)$$

$$d\mu_\theta(\lambda) = f(\lambda)(1 - \theta p_*(\lambda)) d\lambda, \quad -1 \leq \theta \leq 1. \quad (4.4)$$

The notation p_* is reserved for the bounded sine perturbation; the unrelated large slope parameter p will be introduced later.

Theorem 4.1 (A moment-null positive family). *The measures μ_θ are distinct probabilities with common moments*

$$m_n = \frac{\Gamma((n+\beta)/\alpha)}{\Gamma(\beta/\alpha)}. \quad (4.5)$$

These moments, and their formal inverse coefficients, have exact Gevrey order $1/\alpha$.

Proof. The density is nonnegative because $|p_*| \leq 1$. Substitution $u = \lambda^\alpha$ gives the unperturbed moments. The signed moment of the perturbation is

$$\frac{1}{\Gamma(a)} \Im \left[e^{-i\pi\beta} \int_0^\infty u^{(n+\beta)/\alpha-1} e^{-(1-i\tan(\pi\alpha))u} du \right].$$

The complex gamma integral is valid because the real part of its exponential coefficient is positive. Since $1 - i\tan(\pi\alpha) = c^{-1}e^{-i\pi\alpha}$, its phase after multiplication by $e^{-i\pi\beta}$ is πn . The imaginary part is zero. The case $n = 0$ proves normalization. The sine perturbation is not zero almost everywhere, so different parameters give different measures.

Stirling's formula yields $\log m_n = \alpha^{-1}n \log n + O(n)$. The remaining polynomial factors can be absorbed into an exponential constant, so the order $1/\alpha$ is attained, while every smaller order fails. Apply Lemma 3.3 to the inverse. $\square$

Theorem 4.2 (Closed form for the entire transform defect). *For every $t > 0$, the transforms of (4.4) satisfy*

$$S_\theta(t) = S_0(t) + \theta D(t), \quad D(t) = \frac{\pi\alpha}{\Gamma(\beta/\alpha)} t^{-\beta} e^{-At^{-\alpha}} > 0. \quad (4.6)$$

In particular, their common formal expansion omits a nonzero, explicitly known, flat function.

Proof. Let $H(t) = \int f(\lambda)p_*(\lambda)/(1+t\lambda) d\lambda$. After $u = \lambda^\alpha$, introduce $v = (1 - i\tan(\pi\alpha))u$. The phase factors cancel, giving

$$H(t) = \frac{c^a}{\Gamma(a)} \Im \int_{\arg v = -\pi\alpha} \frac{v^{a-1} e^{-v}}{1 - tc^{1/\alpha} v^{1/\alpha}} dv. \quad (4.7)$$

Powers are taken with their principal branches in the wedge $-\pi\alpha \leq \arg v \leq 0$. The original and transformed integrals are absolutely convergent.

Rotate the ray to the positive real axis, passing below the simple pole $r = t^{-\alpha}/c$. There are no other poles in the open wedge: the argument of $v^{1/\alpha}$ lies strictly between $-\pi$ and zero. Large arcs vanish exponentially because $\pi\alpha < \pi/2$, and small arcs vanish because $a > 0$. The integral along the real axis is interpreted as its real principal value plus the lower indentation. The residue is

$$\text{Res}_{v=r} \frac{v^{a-1} e^{-v}}{1 - tc^{1/\alpha} v^{1/\alpha}} = -\alpha r^a e^{-r}. \quad (4.8)$$

The lower indentation, traversed from left to right, contributes $i\pi$ times the residue. All remaining boundary values are real. Thus

$$H(t) = -\frac{\pi\alpha c^a}{\Gamma(a)} r^a e^{-r} = -\frac{\pi\alpha}{\Gamma(a)} t^{-\beta} e^{-At^{-\alpha}}.$$

The definition $1 - \theta p_*$ of the density now gives (4.6). Exponential decay dominates every power of t , proving flatness. $\square$

Example 4.3 (A completely explicit Gevrey-three model). For $\alpha = \beta = 1/3$, one has $m_n = (3n)!$ and

$$S_0(t) = \int_0^\infty \frac{e^{-u}}{1 + tu^3} du, \quad D(t) = \frac{\pi}{3} t^{-1/3} e^{-2t^{-1/3}}. \quad (4.9)$$

Every $\theta \in [-1, 1]$ gives a positive Stieltjes transform $S_0 + \theta D$, and the corresponding inverses share the formal series

$$\widehat{T}(y) = y - 6y^2 + 756y^3 - 376056y^4 + 488904336y^5 + \dots \quad (4.10)$$

The continuum of actual inverse functions cannot be distinguished by computing additional coefficients of this series.

Remark 4.4 (The endpoint and the classical dichotomy). For the unperturbed generalized gamma law with arbitrary $\alpha > 0$, (4.5) has order $1/\alpha$. Thus $\alpha \geq 1/2$ falls within the determinacy criterion, whereas the family above proves indeterminacy for $\alpha < 1/2$. This recovers a classical generalized gamma dichotomy rather than replacing it with a new moment theorem. The contour formula makes an additional quantitative feature visible: its action $A = \sec(\pi\alpha)$ diverges as the boundary is approached from below.

5 Bilateral geometric measures and theta-scale ambiguity

Fix $0 < q < 1$, put $h = -\log q$, and set

$$w_n = q^{n(n-1)/2}, \quad \lambda_n = q^{-n} \quad (n \in \mathbb{Z}). \quad (5.1)$$

The Gaussian decay in n makes every sum used below absolutely convergent. Define

$$M = \sum_{n \text{ even}} w_n = \sum_{n \text{ odd}} w_n, \quad \mu_+ = \frac{1}{M} \sum_{n \text{ even}} w_n \delta_{q^{-n}}, \quad \mu_- = \frac{1}{M} \sum_{n \text{ odd}} w_n \delta_{q^{-n}}. \quad (5.2)$$

The equality defining M follows by pairing n with $1 - n$. Write $S_\pm = S_{\mu_\pm}$, and $S_\eta = (1 - \eta)S_- + \eta S_+$ for $0 \leq \eta \leq 1$. These discrete measures belong to the classical Stieltjes–Wigert moment setting [5, 6, 7].

Proposition 5.1 (Common moments and exact difference equations). *Both measures in (5.2) have moments*

$$m_k = q^{-k(k+1)/2} \quad (k \geq 0). \quad (5.3)$$

Their transforms obey

$$S_+(qt) = 1 - tS_-(t), \quad S_-(qt) = 1 - tS_+(t). \quad (5.4)$$

Consequently every S_η satisfies the same exact equation

$$S_\eta(q^2t) = 1 - qt + qt^2 S_\eta(t). \quad (5.5)$$

The common formal series satisfies $\widehat{S}(qt) = 1 - t\widehat{S}(t)$, although not every member of this particular affine family satisfies that first-order equation exactly.

Proof. The index shift $n = m + k$ gives $w_n q^{-kn} = q^{-k(k+1)/2} w_m$. It preserves or reverses parity according to k ; either parity has normalizing sum M . This proves the moments. For the difference equations, shift $n = m + 1$, use $w_{m+1} = q^m w_m$, and use

$$\frac{q^m}{1 + tq^{-m}} = q^m - \frac{t}{1 + tq^{-m}}.$$

The shifted constant sum is again M , giving (5.4). A second iteration proves (5.5). The formal first-order identity follows directly from (5.3) by comparing coefficients. $\square$

We use the following normalization of the theta function:

$$\Theta_q(z) = \sum_{n \in \mathbb{Z}} q^{n(n-1)/2} z^n, \quad (5.6)$$

$$(a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j),$$

$$\Theta_q(z) = (-z; q)_\infty (-q/z; q)_\infty (q; q)_\infty. \quad (5.7)$$

Equation (5.7) is Jacobi's triple-product identity [10]. It also fixes all normalization constants in what follows.

Theorem 5.2 (Exact reciprocal-theta defect). *For every $t > 0$,*

$$D(t) := S_+(t) - S_-(t) = \frac{(q; q)_\infty^3}{M\Theta_q(t)} = \frac{(q; q)_\infty^2}{M(-t; q)_\infty (-q/t; q)_\infty} > 0. \quad (5.8)$$

In particular,

$$D(qt) = tD(t). \quad (5.9)$$

Proof. Let

$$G(t) = \sum_{n \in \mathbb{Z}} \frac{(-1)^n w_n}{1 + tq^{-n}} = MD(t).$$

The series is normally convergent on compact subsets of $\mathbb{C}^* \setminus \{-q^n : n \in \mathbb{Z}\}$. The same index shift as above gives $G(qt) = tG(t)$, because $\sum (-1)^n q^n w_n = 0$. Also $\Theta_q(qt) = t^{-1} \Theta_q(t)$.

The product $G(t)\Theta_q(t)$ is holomorphic on $\mathbb{C}^*$: the zeros in the triple product cancel the simple poles of G . It is invariant under multiplication by q . It is bounded on a closed fundamental annulus and hence, by this invariance, on all of $\mathbb{C}^*$. Its singularity at zero is removable. Liouville's theorem makes it constant. At $t = -1$, the residue of G is one, whereas $\Theta_q(t)/(1+t) \rightarrow (q; q)_\infty^3$. The constant is therefore $(q; q)_\infty^3$, proving the formula. Every factor in the product form is positive for $t > 0$. $\square$

5.1 A precise log-periodic amplitude

For real u , define

$$P_h(u) = \sum_{n \in \mathbb{Z}} e^{-h(n+u)^2/2}, \quad K_q = \frac{(q; q)_\infty^3}{M}, \quad x = \frac{\log(1/t)}{h}. \quad (5.10)$$

Completing the square in the theta series yields the *exact* identity

$$D(t) = K_q \frac{\exp[-\frac{h}{2}(x - \frac{1}{2})^2]}{P_h(x - \frac{1}{2})}. \quad (5.11)$$

The function P_h is positive, smooth, and one-periodic. Its Fourier coefficients follow by integrating the periodized Gaussian over one period:

$$P_h(u) = \sqrt{\frac{2\pi}{h}} \sum_{k \in \mathbb{Z}} e^{-2\pi^2 k^2 / h} e^{2\pi i k u}. \quad (5.12)$$

The Gaussian Fourier transform justifies this absolutely convergent Fourier expansion; in particular, P_h is not constant.

It follows that D is flat, but

$$\frac{D(t)}{\exp[-\frac{h}{2}(x - \frac{1}{2})^2]} \quad (5.13)$$

has no limit as $t \downarrow 0$. Indeed, choose two phases at which P_h differs and let x tend to infinity along those phases. Thus a single real constant-amplitude monomial at this scale is insufficient. A periodic coefficient, or an equivalent oscillatory extension, is needed. This observation does not exclude every possible extended transseries representation.

Theorem 5.3 (The ambiguity matches the least moment bound). *Let*

$$E_*(t) = \min_{N \in \mathbb{N}} t^N q^{-N(N+1)/2}, \quad \mathbb{N} = \{0, 1, 2, \dots\}. \quad (5.14)$$

For $x \geq 1/2$,

$$E_*(t) = \exp\left[-\frac{h}{2}(x - \frac{1}{2})^2 + \frac{h}{2} \text{dist}(x - \frac{1}{2}, \mathbb{Z})^2\right], \quad (5.15)$$

$$\frac{D(t)}{E_*(t)} = \frac{K_q \exp[-\frac{h}{2} \text{dist}(x - \frac{1}{2}, \mathbb{Z})^2]}{P_h(x - \frac{1}{2})}. \quad (5.16)$$

In particular, $D(t) \asymp_q E_*(t)$ as $t \downarrow 0$: the ratio is bounded above and below by strictly positive constants depending only on q .

Proof. The logarithm of the N -th bound is

$$\log(t^N m_N) = \frac{h}{2}(N - (x - \frac{1}{2}))^2 - \frac{h}{2}(x - \frac{1}{2})^2.$$

For $x \geq 1/2$, a nearest integer to $x - 1/2$ is nonnegative. This proves (5.15). Divide (5.11) by it. The periodic continuous denominator has a positive minimum and finite maximum, and $0 \leq \text{dist}(u, \mathbb{Z}) \leq 1/2$. $\square$

Remark 5.4 (A bound, not an equality of true remainders). The quantity E_* is the best member of the explicit family of bounds in (2.5). The theorem does not assert that it equals the smallest actual truncation error of either transform. It does show that no coefficient-only reconstruction can generally remove ambiguity at that scale. Also, $\log m_n = \frac{h}{2}n^2 + O(n)$, so this example lies beyond every finite Gevrey order.

5.2 Even the exact first-order equation can remain nonunique

One might try to remove the ambiguity by imposing the first-order q -Euler equation exactly. Within the affine family S_η , it selects $\eta = 1/2$. In the full positive-measure class, that conclusion is false.

Proposition 5.5 (Shifted lattices with the same exact equation). *For $r > 0$, define*

$$Z_r = \sum_{n \in \mathbb{Z}} r^{-n} w_n = \Theta_q(1/r), \quad \nu_r = \frac{1}{Z_r} \sum_{n \in \mathbb{Z}} r^{-n} w_n \delta_{rq^{-n}}. \quad (5.17)$$

All these probabilities have moments (5.3). Their transforms $U_r = S_{\nu_r}$ all satisfy

$$U_r(qt) = 1 - tU_r(t). \quad (5.18)$$

If $r/s \notin q^{\mathbb{Z}}$, the transforms and their inverse germs are distinct. Their difference changes sign infinitely often at zero. The corresponding inverse germs also cross infinitely often.

Proof. For the moments, shift $n = m + k$ as in Proposition 5.1; the factors r^k from the support and r^{-k} from the weight cancel. If $a_n = Z_r^{-1} r^{-n} w_n$, then $a_{n+1} = (rq^{-n})^{-1} a_n$. Shifting the transform sum gives

$$U_r(qt) = \sum_n \frac{a_n (rq^{-n})^{-1}}{1 + trq^{-n}} = \sum_n a_n (rq^{-n})^{-1} - tU_r(t) = 1 - tU_r(t).$$

The first sum is $\sum a_{n+1} = 1$.

When $r/s \notin q^{\mathbb{Z}}$, the supports are disjoint, so the measures are different. By Theorem 3.2, so are their transforms. For $V = U_r - U_s$, equation (5.18) gives $V(qt) = -tV(t)$. Choose $t_* > 0$ with $V(t_*) \neq 0$. Iteration shows alternating signs at $q^n t_*$; continuity gives a zero between successive such points.

For inverse transport, write $y = t/U_s(t)$, so $T_s(y) = t$. The residual for the other inverse is $t - yU_r(t) = -yV(t)$. Since its residual is strictly increasing, $\text{sgn}(T_r(y) - T_s(y)) = \text{sgn} V(t)$. The sequence $t = q^n t_*$ tends to zero and so does y ; its signs alternate. This proves infinitely many inverse crossings. $\square$

Corollary 5.6 (A simultaneous germ-field obstruction). *The two inverse germs in Proposition 5.5 cannot both lie in a field of actual real-valued germs at zero. In particular, after $y = 1/x$, they cannot both belong to one Hardy field at infinity.*

Proof. Their difference is a nonzero germ but has zeros arbitrarily close to zero. It therefore has no reciprocal germ. A field cannot contain such a nonzero element. $\square$

Remark 5.7. This is a statement about simultaneous realization, not a proof that either inverse individually lies in no Hardy field. The basic moment-indeterminate lattice families are classical. Their role here is to show exactly which extra analytic equations still fail to turn formal data into unique inverse data.

6 Nonlinear transport and an unavoidable error floor

In this section $S_a = S_{a_0} + (a - a_0)D$ denotes either the gamma family with physical interval $[-1, 1]$, or the ordered theta family with physical interval $[0, 1]$. The word *physical* means simply that the associated measure is positive; it does not refer to an external physical model. In both families $D(t) > 0$, and every member has the same moments.

Lemma 6.1 (Exact derivative with respect to the hidden parameter). *For $y > 0$ and a physical parameter a ,*

$$\partial_a T_a(y) = \frac{yD(T_a(y))}{1 - yS'_a(T_a(y))} > 0. \quad (6.1)$$

For two physical parameters a, b ,

$$T_a(y) - T_b(y) = \int_b^a \frac{yD(T_\theta(y))}{1 - yS'_\theta(T_\theta(y))} d\theta. \quad (6.2)$$

Every finite Taylor expansion of T_a at zero is independent of a .

Proof. Differentiate $T_a = yS_a(T_a)$. The denominator is at least one by positivity of the measure, proving the formula and its sign. Integration gives the second statement. Parameter endpoints are handled by one-sided derivatives or by local analytic continuation. Shared Taylor coefficients follow from Theorem 3.2. $\square$

Theorem 6.2 (Two quantitative inverse-gap laws). *The following expansions hold uniformly for distinct physical parameters a, b , including in the limit $a - b \rightarrow 0$.*

For the gamma family, with the parameters of (4.1),

$$\frac{T_a(y) - T_b(y)}{(a - b)yD(y)} = 1 - \alpha Am_1 y^{1-\alpha} + (\beta - 1)m_1 y + O(y^{2-2\alpha}). \quad (6.3)$$

For the theta family, define the bounded one-periodic function

$$a_q(x) = -\frac{1}{2} + \frac{P'_h(x - \frac{1}{2})}{hP_h(x - \frac{1}{2})}. \quad (6.4)$$

With $x = \log(1/y)/h$ and $m_1 = q^{-1}$,

$$\frac{T_a(y) - T_b(y)}{(a - b)yD(y)} = 1 - m_1 y(x + a_q(x) + 1) + O(y^2(1 + x)^2). \quad (6.5)$$

Proof. The shared finite moments give, uniformly in the physical parameter,

$$T_a(y) = y - m_1 y^2 + O(y^3), \quad (1 - yS'_a(T_a(y)))^{-1} = 1 - m_1 y + O(y^2).$$

Uniformity follows either from bounded densities relative to a common base in the gamma case, or from convexity in the theta case, and the finite-moment derivative bounds.

For gamma, expand the logarithm of the exact defect:

$$\begin{aligned} \log \frac{D(T_a(y))}{D(y)} &= -\beta \log \frac{T_a(y)}{y} - Ay^{-\alpha} \left[\left(\frac{T_a(y)}{y} \right)^{-\alpha} - 1 \right] \\ &= -\alpha Am_1 y^{1-\alpha} + \beta m_1 y + O(y^{2-\alpha}). \end{aligned}$$

Exponentiating introduces an error of order $y^{2-2\alpha}$. Multiplying by the denominator expansion gives the integrand in (6.2), divided by $yD(y)$. The asserted gamma law follows. Because $\alpha < 1/2$, the error has order strictly greater than one, so the displayed linear term is informative.

For theta, direct differentiation of (5.11) gives

$$t \frac{D'(t)}{D(t)} = x + a_q(x), \quad x = \frac{\log(1/t)}{h}. \quad (6.6)$$

Since a_q and all its real derivatives are bounded, integration over $\log(T_a(y)/y) = -m_1 y + O(y^2)$ yields

$$\log \frac{D(T_a(y))}{D(y)} = -m_1 y(x + a_q(x)) + O(y^2(1 + x)).$$

Exponentiate, multiply by the denominator factor, and use (6.2). All estimates are uniform in a , hence also in the averaged interval from b to a . $\square$

For the cubic example, equation (6.3) becomes particularly simple:

$$\frac{T_a(y) - T_b(y)}{(a - b)yD(y)} = 1 - 4y^{2/3} - 4y + O(y^{4/3}). \quad (6.7)$$

Thus a flat perturbation does not merely persist: its first nonlinear relative correction can be computed without resolving the full coefficient expansion.

Theorem 6.3 (A coefficient-only information floor). *Let an estimator $\mathcal{A}(m, y)$ depend only on y and the full common moment sequence, or equivalently on the full common inverse coefficient sequence. For any two physical members,*

$$\max\{|\mathcal{A} - T_a(y)|, |\mathcal{A} - T_b(y)|\} \geq \frac{1}{2}|T_a(y) - T_b(y)|. \quad (6.8)$$

For the theta endpoints, this lower bound is asymptotic to $yD(y)/2$ and hence is bounded below by $c_q y E_(y)$ for all sufficiently small y , with $c_q > 0$. For the gamma endpoints $a = 1, b = -1$, it is asymptotic to $yD(y)$.*

Proof. The estimator receives identical input for both members. The triangle inequality gives (6.8). Apply Theorem 6.2, and for theta use Theorem 5.3. $\square$

Remark 6.4. This lower bound concerns a worst-case error over a family of admissible realizations. It is not a limit on methods supplied with the actual measure, an exact integral, a summation prescription, or another datum that fixes the hidden parameter. In the gamma case, we do not identify the defect action A with the least-moment-bound action: they are different quantities.

7 Convergent expansions in the flat ambiguity parameter

A finite truncation of the common formal power series cannot describe the ambiguity in Section 6. A different expansion can: keep one actual realization as the baseline, and expand in the hidden parameter. In this section the letters a, a_0 denote family parameters, not the gamma quotient β/α .

Fix a physical baseline a_0 , a small $y > 0$, and set

$$t_0 = T_{a_0}(y), \quad S = S_{a_0}, \quad H(t) = t - yS(t), \quad B_0 = 1 - yS'(t_0), \quad D_0 = D(t_0). \quad (7.1)$$

The deformed inverse equation is

$$H(t) = \delta y D(t), \quad \delta = a - a_0. \quad (7.2)$$

Complex δ is allowed in the local analytic discussion, even though only a bounded real interval corresponds to positive measures.

Theorem 7.1 (Exact sector coefficients and an explicit tail bound). *Assume $ym_1 \leq 1/2$, and choose $0 < r < t_0/2$. On a neighborhood of the closed disc $|t - t_0| \leq r$, define the removable analytic quotient*

$$A_y(t) = \frac{t - t_0}{H(t)}, \quad A_y(t_0) = B_0^{-1}, \quad \phi_y(t) = yD(t)A_y(t). \quad (7.3)$$

Let M_D be any bound for $|D|$ on the disc and set $M = 2yM_D$. For $\rho = |\delta|M/r < 1$, the branch through t_0 is represented by an absolutely convergent series

$$T_{a_0+\delta}(y) = t_0 + \sum_{k \geq 1} \delta^k A_k(y), \quad (7.4)$$

$$A_k(y) = \frac{1}{k!} \left. \frac{d^{k-1}}{dt^{k-1}} (\phi_y(t)^k) \right|_{t=t_0}. \quad (7.5)$$

Its coefficients and tails satisfy

$$|\delta^k A_k(y)| \leq \frac{r}{k} \rho^k, \quad (7.6)$$

$$\left| T_{a_0+\delta}(y) - t_0 - \sum_{k=1}^K \delta^k A_k(y) \right| \leq \frac{r \rho^{K+1}}{(K+1)(1-\rho)}. \quad (7.7)$$

For physical $a_0 + \delta$, this branch is the positive inverse already defined in Section 2.

Proof. The disc lies in the right half-plane. There $|S'(t)| \leq m_1$, by (2.3). Since $H(t_0) = 0$,

$$\frac{H(t)}{t - t_0} = 1 - y \int_0^1 S'(t_0 + s(t - t_0)) ds.$$

The right side differs from one by at most $ym_1 \leq 1/2$. Thus it is nonzero and $|A_y| \leq 2$ throughout the disc. Equation (7.2) is equivalent there to $t - t_0 = \delta \phi_y(t)$.

On the boundary, $|\delta \phi_y| \leq |\delta|M < r = |t - t_0|$. Rouché's theorem gives one root, counted with multiplicity, in the disc. It is simple and depends holomorphically on δ for $|\delta| < r/M$. For real physical parameters this root is real by conjugation symmetry, and positive because $r < t_0$. Uniqueness of the positive inverse identifies it with $T_{a_0+\delta}$.

The analytic Lagrange formula gives (7.5). For completeness, putting $w = t - t_0$, the coefficient is $k^{-1}[w^{k-1}] \phi_y(t_0 + w)^k$. Cauchy's coefficient estimate on $|w| = r$ gives $|A_k| \leq M^k/(kr^{k-1})$. Summing these bounds and using $1/k \leq 1/(K+1)$ for $k \geq K+1$ proves (7.7). $\square$

Proposition 7.2 (The first two exact sectors). *With the data of (7.1),*

$$A_1(y) = \frac{yD_0}{B_0}, \quad (7.8)$$

$$A_2(y) = \frac{y^2 D_0 D'(t_0)}{B_0^2} + \frac{y^3 D_0^2 S''(t_0)}{2B_0^3}. \quad (7.9)$$

Proof. Expand $H(t_0 + w) = B_0 w - y S''(t_0) w^2/2 + O(w^3)$. Hence $A_y(t_0) = B_0^{-1}$ and $A'_y(t_0) = y S''(t_0)/(2B_0^2)$. Substitution in (7.5), or direct implicit differentiation twice, gives the formulas. $\square$

The coefficients contain successively higher powers of D_0 , multiplied by slope factors. Thus (7.4) resolves the beyond-all-orders ambiguity rather than merely recomputing the common power series. It is an expansion *around an actual baseline*; it does not supply a canonical choice of that baseline from its formal jet.

7.1 Explicit discs in the two families

The next bounds give completely specified sufficient convergence conditions. They are deliberately simple, rather than asymptotically optimal. Section 8 will determine the leading optimal radius.

Proposition 7.3 (A gamma disc). *For the gamma defect, put*

$$p = \alpha A t_0^{-\alpha}, \quad r = \frac{t_0}{8(1 + \beta + p)}. \quad (7.10)$$

Then $M_D = e^{3/8} D_0$ is valid in Theorem 7.1. For all sufficiently small y ,

$$\frac{M}{r} \leq 32e^{3/8}(1 + \beta + p)D_0 \rightarrow 0. \quad (7.11)$$

Proof. On $|t - t_0| \leq t_0/2$, using the principal powers,

$$\left| \frac{D'(t)}{D(t)} \right| \leq \frac{2\beta + 2^{1+\alpha}p}{t_0} \leq \frac{2\beta + 3p}{t_0}.$$

The nonvanishing explicit defect has a holomorphic logarithm on this disc. Integrating its logarithmic derivative along a straight segment of length at most r bounds the logarithmic variation by $3/8$. This proves the bound for M_D . Since $t_0/y \rightarrow 1$, eventually $y/t_0 \leq 2$; substituting $M = 2yM_D$ gives (7.11). Its limit is zero by exponential flatness. $\square$

Proposition 7.4 (A theta disc). *For the theta defect, let $t_0 < 1$, and put*

$$x_0 = \frac{\log(1/t_0)}{h}, \quad B = x_0 + \frac{4}{1-q}, \quad r = \frac{t_0}{8B}. \quad (7.12)$$

Then $M_D = e^{1/4} D_0$ is valid in Theorem 7.1, and eventually

$$\frac{M}{r} \leq 32e^{1/4}BD_0 \rightarrow 0. \quad (7.13)$$

Proof. Logarithmically differentiate the product in (5.8):

$$t \frac{D'(t)}{D(t)} = - \sum_{n \geq 0} \frac{tq^n}{1 + tq^n} + \sum_{n \geq 0} \frac{q^{n+1}}{t + q^{n+1}}.$$

For $|t - t_0| \leq t_0/2$, the first absolute sum is at most $3/[2(1 - q)]$. In the second, the terms with $n + 1 \leq x_0$ are each at most one, while the remaining sum is at most $2/(1 - q)$. Thus $|tD'/D| \leq B$, and $|D'/D| \leq 2B/t_0$. No zero or pole of D lies in this disc. Integrating its logarithmic derivative over radius r gives variation at most $1/4$. The last estimate follows from $y/t_0 \leq 2$ and the log-Gaussian flatness of D_0 . $\square$

Both propositions imply convergence uniformly over every bounded physical parameter interval when y is sufficiently small. Their bounds are also effective once the actual baseline and its defect are numerically enclosed. The floating-point examples later in the article do not by themselves provide such outward-rounded enclosures.

8 A universal Lambert– W limit and the parameter radius

The scale of the sharp parameter radius is larger than the elementary Cauchy-disc radius by a bounded factor. More importantly, it has a universal limit. Continue to fix a physical baseline a_0 , and use (7.1). Define

$$p = \begin{cases} \alpha A t_0^{-\alpha}, & \text{gamma,} \\ \log(1/t_0)/h, & \text{theta,} \end{cases} \quad v = \frac{p(t - t_0)}{t_0}, \quad \kappa = \delta \frac{pyD_0}{t_0}. \quad (8.1)$$

In both cases $p \rightarrow \infty$ as $y \downarrow 0$. Equation (7.2) becomes

$$\kappa = \Psi_y(v) := \frac{pH(t_0(1 + v/p))}{t_0} \frac{D_0}{D(t_0(1 + v/p))}. \quad (8.2)$$

On every fixed bounded complex v -set, this formula is holomorphic for all sufficiently small y . It no longer involves a subtraction of two unspecified formal series.

Lemma 8.1 (Uniform local convergence of the deformation map). *For either family,*

$$\Psi_y(v) \longrightarrow v e^{-v} \quad (8.3)$$

locally uniformly in $v \in \mathbb{C}$, with convergence of every derivative.

Proof. Since $H(t_0) = 0$ and $|S'| \leq m_1$ in the right half-plane,

$$\frac{pH(t_0(1 + v/p))}{t_0} = v + O_R(y) \quad (|v| \leq R).$$

For gamma, the explicit logarithm gives

$$\log \frac{D(t_0(1 + v/p))}{D_0} = -\beta \log(1 + v/p) - \frac{p}{\alpha} ((1 + v/p)^{-\alpha} - 1) = v + O_R(p^{-1}).$$

For theta, integrate (6.6) over $\log(1 + v/p)$. The periodic factor and its derivatives are uniformly bounded in a fixed complex strip about the real phase axis: positivity on that axis and compactness of one period supply a zero-free strip. The resulting logarithm is again $v + O_R(p^{-1})$. This proves (8.3). Local uniform holomorphic convergence implies convergence of all derivatives by Cauchy's formula. $\square$

Theorem 8.2 (Cayley coefficients, a real fold, and a sharp leading radius). *Let R_y be the radius of the Taylor series in δ for the branch $T_{a_0+\delta}(y)$ through t_0 . Then*

$$R_y \sim \frac{t_0}{e p y D_0}. \quad (8.4)$$

In scaled coordinates, the branch converges locally uniformly for $|\kappa| < 1/e$ to

$$v_0(\kappa) = -W_0(-\kappa) = \sum_{k \geq 1} \frac{k^{k-1}}{k!} \kappa^k, \quad v_0 e^{-v_0} = \kappa. \quad (8.5)$$

Consequently, for every fixed $k \geq 1$,

$$A_k(y) = \frac{t_0}{p} \left(\frac{pyD_0}{t_0} \right)^k \left(\frac{k^{k-1}}{k!} + O_k(p^{-1}) \right). \quad (8.6)$$

There is a real critical point $v_c(y) \rightarrow 1$ of Ψ_y , with critical value $\kappa_c(y) \rightarrow 1/e$, on the continuation of the branch from the origin. It is a nondegenerate square-root fold.

Proof. We separate the lower radius bound from the upper bound; the latter cannot be inferred merely from coefficient limits at fixed k .

Analytic continuation inside every smaller disc. Fix $r < 1/e$ and choose $r < r_1 < 1/e$. The inverse $v_0 = -W_0(-\kappa)$, defined equivalently by its Taylor series, is holomorphic on a neighborhood of $|\kappa| \leq r_1$. Its image is compact, and $(ve^{-v})' = e^{-v}(1-v)$ is nonzero there. Uniform local inverse neighborhoods can therefore be chosen around $v_0(\kappa)$, with one common sufficiently small radius on the compact κ -disc. Taylor's theorem and the positive minimum of the derivative give a uniform nonzero lower bound for $|ve^{-v} - \kappa|$ on the boundaries of those neighborhoods. Lemma 8.1 and Rouché's theorem then give a unique nearby root of $\Psi_y(v) = \kappa$, analytic in κ . Local uniqueness makes these roots agree on overlapping neighborhoods, producing a single analytic branch on $|\kappa| \leq r$. The roots converge uniformly to v_0 . The same argument, using the $O(p^{-1})$ error in the proof of the lemma, gives that convergence rate on every smaller fixed disc. Hence the scaled Taylor radius has liminf at least $1/e$.

A singularity on the same branch. The derivative of ve^{-v} has a simple zero at $v = 1$, and its second derivative there is $-1/e$. Holomorphic convergence gives one simple zero $v_c(y)$ of Ψ'_y near one. Reality of the functions implies that this zero is real. On every interval $[0, 1 - \varepsilon]$, Ψ'_y is positive for sufficiently small y . In a fixed small neighborhood of one, Ψ''_y is negative, so Ψ'_y decreases through its unique zero. It follows that the branch from zero increases to v_c , at $\kappa_c = \Psi_y(v_c) \rightarrow 1/e$.

The Taylor expansion at the critical point is $\Psi_y(v) = \kappa_c + \Psi''_y(v_c)(v - v_c)^2/2 + O((v - v_c)^3)$. Since $\Psi''_y(v_c) \neq 0$, the inverse has a genuine square-root singularity there. An analytic continuation of the Taylor series across that point on this branch is impossible. Thus the scaled radius has limsup at most $1/e$. Combining both bounds and undoing (8.1) proves (8.4).

Finally, the inverse of $v \mapsto ve^{-v}$ has the coefficients in (8.5) by Lagrange inversion. Uniform convergence of the scaled analytic inverses on a fixed circle gives convergence of coefficients, with error $O_k(p^{-1})$. Rescaling yields (8.6). $\square$

Remark 8.3 (What the radius theorem does and does not locate). The theorem determines the leading asymptotic Taylor radius. It does not assert that the real fold is the exact nearest complex singularity for every fixed nonzero y . Competing singularities could alter subleading information about that radius. The fold location itself, which is a local real statement, is calculated more precisely next. For small y , its δ -value tends to infinity, so it lies far outside any fixed positive-measure parameter interval. No loss of existence of the physical positive inverse is being asserted.

8.1 The first fold correction and the surviving theta phase

Lemma 8.4 (A two-term scaled defect expansion). *Uniformly on bounded complex v -sets,*

$$\log \frac{D(t_0(1 + v/p))}{D_0} = v + \frac{Q_y(v)}{p} + O_R(p^{-2}), \quad (8.7)$$

where

$$Q_y(v) = \begin{cases} -\beta v - \frac{1+\alpha}{2}v^2, & \text{gamma,} \\ a_q(p)v - \frac{1}{2}v^2, & \text{theta.} \end{cases} \quad (8.8)$$

Moreover,

$$\Psi_y(v) = ve^{-v} \left(1 - \frac{Q_y(v)}{p} \right) + O_R(p^{-2}), \quad (8.9)$$

with the same control for finitely many derivatives on smaller compact sets.

Proof. The gamma formula follows by taking two further terms in the binomial and logarithmic expansions used in Lemma 8.1. For theta, put $s = \log(t/t_0)$, so the phase is $p - s/h$. Integrate

$$\frac{d}{ds} \log D(t_0 e^s) = p - s/h + a_q(p - s/h)$$

from zero to $\log(1 + v/p)$. The terms of orders one and p^{-1} are exactly those in (8.8). Bounded complex-strip derivatives of a_q control the remainder uniformly in its phase.

The residual prefactor in (8.2) is $v + O_R(y)$. In the gamma case $y \asymp t_0 \asymp p^{-1/\alpha} = o(p^{-2})$, because $\alpha < 1/2$. In the theta case $y \asymp e^{-hp} = o(p^{-2})$. The residual correction is therefore smaller than the stated error. Exponentiate (8.7) with a minus sign and multiply by this prefactor. Holomorphy again gives derivative control. $\square$

Theorem 8.5 (Explicit gamma and theta fold corrections). *The real critical point and value from Theorem 8.2 satisfy*

$$v_c = 1 - \frac{Q'_y(1)}{p} + O(p^{-2}), \quad \kappa_c = e^{-1} \left(1 - \frac{Q_y(1)}{p} \right) + O(p^{-2}). \quad (8.10)$$

Thus, in the gamma case,

$$v_c = 1 + \frac{\beta + 1 + \alpha}{p} + O(p^{-2}), \quad (8.11)$$

$$\kappa_c = e^{-1} \left(1 + \frac{\beta + (1 + \alpha)/2}{p} \right) + O(p^{-2}). \quad (8.12)$$

In the theta case,

$$v_c = 1 + \frac{1 - a_q(p)}{p} + O(p^{-2}), \quad (8.13)$$

$$\kappa_c = e^{-1} \left(1 + \frac{1/2 - a_q(p)}{p} \right) + O(p^{-2}). \quad (8.14)$$

The errors are uniform in the theta phase for fixed q . Locally on the two inverse branches,

$$v = v_c \pm \sqrt{\frac{2(\kappa_c - \kappa)}{-\Psi''_y(v_c)}} + O(\kappa_c - \kappa), \quad -\Psi''_y(v_c) \rightarrow e^{-1}. \quad (8.15)$$

Proof. Differentiate (8.9) near $v = 1$. Writing $v_c = 1 + d/p + O(p^{-2})$, the coefficient at order p^{-1} in $\Psi'_y(v_c) = 0$ is $-e^{-1}(d + Q'_y(1))$. This proves the first part of (8.10). The derivative of ve^{-v} vanishes at one, so the displacement of the critical point contributes only at order p^{-2} to the critical value. Evaluation of (8.9) at one proves the second part. Substitute (8.8) for the displayed specializations. Finally, invert the quadratic Taylor term at the nondegenerate critical point to obtain (8.15). $\square$

The periodic coefficient is visible in both the inverse-gap law (6.5) and the singularity law (8.14). Thus nonlinear inversion does not wash out the phase of the flat ambiguity. The leading normal form is universal, but its first correction remembers the realization scale.

Corollary 8.6 (An abstract local criterion for the same radius law). *The leading radius and Cayley-coefficient conclusions do not require a moment model. Suppose a family of local equations has the form $H_y(t) = \delta y D_y(t)$, with $H_y(t_0) = 0$, $D_y(t_0) > 0$, and a scale $p_y \rightarrow \infty$. Suppose its data are real on the positive axis and holomorphic on the expanding scaled discs, and, locally uniformly in complex v ,*

$$\frac{p_y H_y(t_0(1 + v/p_y))}{t_0} \rightarrow v, \quad \frac{D_y(t_0(1 + v/p_y))}{D_y(t_0)} \rightarrow e^v. \quad (8.16)$$

Then the branch through t_0 has parameter radius $R_y \sim t_0/(e p_y y D_y(t_0))$, a real fold on its continuation, and normalized fixed-order coefficient limits $k^{k-1}/k!$. If the convergence in (8.16) has rate $O(p_y^{-1})$, the coefficient errors have that rate as well.

Proof. The scaled deformation map is the quotient of the two expressions in (8.16). It converges holomorphically to ve^{-v} . The lower-radius continuation, local real critical point, upper-radius obstruction, and coefficient extraction in the proof of Theorem 8.2 apply without change. $\square$

9 Reproducible computations and verification boundaries

The accompanying program `verify.py` uses exact integer arithmetic for the cubic inverse coefficients and arbitrary-precision arithmetic for finite analytic consistency tests. The recorded run used Python 3.13.5, mpmath 1.3.0, and 130 decimal digits. All assertions in that run passed. The exact environment and results are recorded in `verification_results.json`; there are no network dependencies.

9.1 Exact arithmetic and independent transform evaluations

The program computes the first twelve coefficients for $m_n = (3n)!$ using the finite coefficient formula and independently substitutes them into the defining formal equation. The first five are displayed in (4.10). For the theta family at $q = 1/2$, moments through order twelve are checked separately on the even and odd lattices. Transform defects are evaluated both as direct differences of Stieltjes sums and from the theta formulas. The exact q^2 equation and the defect scaling are tested independently. The shifted-lattice first-order equation and its alternating difference are also tested.

For the gamma defect, direct quadrature of the signed integral in the cubic case gives the values in Table 1. The test compares these with the negative of the closed form in (4.9).

Table 1: Cubic signed Stieltjes defect, checked against the exact formula.

t	$\int_0^\infty \frac{e^{-u} \sin(\sqrt{3}u - \pi/3)}{1 + tu^3} du = -D(t)$
1	-0.141722777195878776146374
0.1	-0.0303419687206032999616225
0.01	-0.000451920923529153552470793

9.2 Inverse transport and the fold

Table 2 compares the actual endpoint inverse gap with its leading scale and first relative correction at $q = 1/2$. At integer phases $x = N$, symmetry gives $P'_h(N - 1/2) = 0$, hence $a_q(N) = -1/2$. The final column is therefore $1 - 2y(N + 1/2)$. The value at $N = 4$ is not in a particularly small asymptotic regime; it is included to avoid displaying only favorable large- N cases.

At these integer phases, $D(y)/E_*(y) = 0.004468440172262834\dots$. The phase is fixed, so this repeated ratio illustrates the periodic formula rather than a general phase-independent constant. For the cubic model, the measured ratio at $y = 10^{-4}$ is $0.991133461317105922\dots$, compared with the first corrected value $0.990982261239872465\dots$ from (6.7).

Table 2: Theta-family inverse gap, $q = 1/2$ and $y = 2^{-N}$.

N	$D(y)$	$(T_+ - T_-)/(yD(y))$	First correction
4	$6.98194 \cdot 10^{-5}$	0.644450288768	0.437500000000
8	$1.66462 \cdot 10^{-11}$	0.936987457980	0.933593750000
12	$6.05587 \cdot 10^{-23}$	0.993922761611	0.993896484375
20	$2.84745 \cdot 10^{-60}$	0.999960900309	0.999960899353

To test the fold, the program fixes $t_0 = 2^{-p}$ and computes $y = t_0/S_-(t_0)$. It evaluates the residual by subtracting the integrands before summation, rather than subtracting two nearly equal order-one numbers. The critical equation is

$$H'(t)D(t) - H(t)D'(t) = 0. \quad (9.1)$$

Table 3 compares the resulting critical values with (8.14). At these phases the predictor is $e^{-1}(1 + 1/p)$. The bounded last column is consistent with the proved $O(p^{-2})$ remainder.

Table 3: The real theta fold and its first correction.

p	v_c	κ_c	$e^{-1}(1 + 1/p)$	p^2 times error
12	1.15750238455	0.404091437651	0.398536061269	0.7999741990
20	1.08555562628	0.388065092191	0.386273413230	0.7166715845
32	1.05078622850	0.380048631028	0.379375673708	0.6891082956

Finally, the exact first two parameter sectors at $q = 1/2, y = 2^{-8}$ are approximately

$$A_1 = 6.09270122025727 \cdot 10^{-14}, \quad A_2 = 7.19277265023981 \cdot 10^{-24}.$$

For the endpoint displacement $\delta = 1$, the observed omitted tail is approximately $1.20633833887303 \cdot 10^{-33}$; the explicit bound (7.7), evaluated with the theta disc, is approximately $1.42599129686661 \cdot 10^{-30}$. This verifies that the implemented normalizations and the stated analytic estimate are mutually consistent in this test case.

9.3 What these computations certify

The integer coefficient identities are exact finite calculations. All quadratures, infinite-sum truncations, root computations, and theta evaluations in the program use high-precision floating-point arithmetic, not outward-rounded interval arithmetic. Thus the displayed decimals are consistency evidence, not independently certified numerical endpoints. The program does not prove the infinite identities, the asymptotic theorems, or the absence of additional complex singularities. Those assertions rely on the proofs above. A future interval implementation would need explicit enclosures for every tail, quadrature, and rounded arithmetic operation.

10 Consequences for a transseries realization architecture

10.1 Three objects that must not be identified

The examples distinguish three different kinds of data. The first is a formal coefficient sequence, or formal inverse $\widehat{T}$. The second is a selected analytic representative T_{a_0} . The third is the class of all admissible representatives and its flat deformations. A symbolic inversion algorithm acts

on the first object. An evaluated residual certificate acts on the second. A theorem such as (6.8) or (8.4) describes the third. Confusing these objects turns a correct formal computation into an incorrect claim of analytic uniqueness.

Positive measures provide a particularly useful test case because they supply complete monotonicity, real-root uniqueness, and signed remainder bounds, yet still leave nontrivial analytic moduli above the determinacy threshold. The models therefore rule out the proposed implication “positive Stieltjes structure plus all inverse coefficients implies a unique analytic inverse,” while identifying exactly when the implication becomes valid.

10.2 A natural representation of the additional data

In the gamma example, a representation can retain an actual baseline and a coefficient multiplying the generator $t^{-\beta}e^{-At^{-\alpha}}$. Nonlinear inversion generates successive powers of that generator and its logarithmic derivatives. The convergent parameter series gives a direct analytic meaning to these extra sectors without assigning a sum to the divergent common power series.

For theta, retaining only a constant coefficient at the log-Gaussian scale loses information. A representation must also retain the phase $x \bmod 1$, or the periodic function $1/P_h(x - 1/2)$. Its Fourier expansion produces complex oscillatory factors, so a system restricted to purely real nonoscillatory monomials should not silently treat it as an ordinary one-term asymptotic equivalent. The fold correction shows that this is not a superficial formatting issue: the phase affects the location of an actual analytic singularity.

These observations are compatible with the repository’s distinction between formal well-based supports, flat-function interfaces, and concrete analytic realization [2, 3]. They are not a replacement for proving the missing general realization theorems.

10.3 A realistic formalization boundary

A staged formalization would begin with finite algebra and order facts: the weak-composition coefficient identity, its sign, the factorial bound, and the monotonicity/residual certificate. The moment-determinacy bridge then needs measure push-forwards, polynomial approximation on a compact interval, and uniqueness of Fourier transforms under exponential integrability. The gamma and theta examples require complex integration or equivalent special-function identities. The radius theorem requires uniform holomorphic implicit inversion and a carefully stated local fold lemma.

Existing repository modules on flatness, well-based supports, and asymptotic scales are relevant interfaces, but their presence alone does not certify these arguments. This report supplies no Lean source and makes no claim that those dependencies have already been formalized in the required form. The finite Python checks likewise are not a substitute for a proof assistant.

Editorial note (ProveIt, 2026-09-29). The modules meant are `Analysis/Transseries/Lean/Transseries/TransseriesFlat.lean`, `TransseriesWellBased.lean` and `TransseriesScale.lean` in the same directory. Two machine-checked facts are related but not relied on: `Fabius.exists_eq_in_residual_interval` (`Analysis/FabiusFunction/Lean/FabiusFunction/MeanValueBracket.lean`), the monotone residual bracket behind (2.7) in Proposition 2.3, and `Fabius.cayleyTree` (`Analysis/FabiusFunction/Lean/FabiusFunction/CayleyTreeFunction.lean`), the real function $-W_0(-w)$ for $w \leq e^{-1}$, without its power series. No statement of this article is formalized.

11 Further research questions

Q1. Determine the full inverse ambiguity envelope. For a fixed indeterminate Stieltjes moment sequence, determine $\inf_{\mu} T_{\mu}(y)$ and $\sup_{\mu} T_{\mu}(y)$ over all representing probabilities, not only over the constructed affine families. Translate the full moment parametrization into sharp nonlinear envelopes. Are the even and odd lattice inverses asymptotically extremal at the least-bound scale, or is the constant in the information floor strictly improvable? No extremality claim for those two measures has been proved here.

Q2. Locate the exact nearest deformation singularities. Theorem 8.2 gives the leading radius, and Theorem 8.5 locates the real fold to one further order. Determine whether that fold is the nearest singularity for all sufficiently small y . If not, classify the competing complex singularities and their phase dependence. This is necessary before identifying the second-order expansion of the Taylor radius with the second-order expansion of the real fold.

Q3. Develop joint critical limits. The results keep α, β, q fixed. Study the joint limit $\alpha \uparrow 1/2, y \downarrow 0$, where $A = \sec(\pi\alpha)$ diverges and moment determinacy reappears at the endpoint. Separately, analyze $q \uparrow 1, y \downarrow 0$, where the theta oscillation is exponentially small in $1/h$. A uniform answer must distinguish small amplitude from exact absence of a phase.

Q4. Relate positive realizations to canonical summation. For the exact q -Euler family in Proposition 5.5, identify which measures, if any, coincide with particular analytic q -summation prescriptions. Determine the additional sectorial growth or boundary data that restore uniqueness, and then prove which of those selections commute with the nonlinear map $S \mapsto T$. The different summation procedures discussed in [8] make this a question about specified hypotheses, not merely nomenclature.

Q5. Reconstruct the hidden parameter from minimal extra data. For the ordered one-parameter families, one exact transform value fixes the parameter. Quantify the conditioning of this reconstruction when that value is known only approximately. Combine the information floor with the exact derivative (6.1) to identify optimal measurement scales. For the full moment-indeterminate family, determine which finite or countable supplementary observations identify a measure and its inverse.

Q6. Extend the radius criterion beyond the two models. Find verifiable conditions on logarithmic derivatives of a flat defect that imply the complex-uniform limits in Corollary 8.6. For example, formulate conditions for regularly varying actions and for several competing flat scales. Determine when one still obtains a single Lambert fold and when several scales interact to produce a different limiting implicit equation.

Q7. Build an algebra of periodic-coefficient sectors. Construct a precise analytic or formal framework in which the log-periodic coefficient $1/P_h$, its derivatives, products, and inverse-sector expansions are legitimate objects. Specify the support and convergence conditions required for composition. Clarify the boundary between a usable oscillatory calculus and a realization in an ordered Hardy field; Proposition 5.5 provides an explicit test that any simultaneous-realization claim must pass.

Q8. Formalize the positive-measure bridge and certify numerics. A concrete first milestone is a formal proof of Lemma 3.3 and the implication “formal inverse coefficient bound $\Rightarrow$ moment bound.” A second is the exact identifiability theorem. Independently, implement outward-rounded versions of the theta and gamma evaluations and use (2.7) and (7.7) to produce fully rigorous inverse and sector enclosures. These are separate tasks from a full formalization of the contour and fold arguments.

12 Conclusion

The positive Stieltjes inverse problem gives a complete, sharply scoped answer to a realization question relevant to the ProveIt transseries program. Formal inversion neither creates nor removes the identifiability obstruction: that obstruction is exactly the underlying moment problem. A sign-definite coefficient formula transfers the classical order-two determinacy threshold to the nonlinear inverse. Beyond it, exact exponential and theta defects expose the missing analytic data.

The ambiguity is not an arbitrary nuisance term. It has an exact parameter derivative, a convergent nonlinear sector expansion, an information-theoretic error floor, and a universal rescaled deformation map. The leading map ve^{-v} determines the asymptotic parameter radius and Cayley coefficients, while the first fold correction retains the specific gamma or theta structure. These conclusions supply explicit theorems and counterexamples on which a broader realization and certified-inversion theory can build.

A The Lagrange identity used in both coefficient calculations

We record a short proof to make clear that the two uses of Lagrange inversion in the paper have the same normalization. Let $S(t)$ be a formal series with $S(0) = 1$, and let $T = yS(T)$. Formal substitution $y = t/S(t)$ in the residue formula for $[y^n]T$ gives

$$\begin{aligned} [y^n]T &= \text{res}_{t=0} \left(t^{-n} S(t)^n - t^{-(n-1)} S(t)^{n-1} S'(t) \right) dt \\ &= [t^{n-1}]S(t)^n - \frac{1}{n} [t^{n-2}] \frac{d}{dt} S(t)^n \\ &= \frac{1}{n} [t^{n-1}]S(t)^n. \end{aligned}$$

For $n = 1$, the coefficient with negative index is zero. For all n , the second term uses $[t^{n-2}](S^n)' = (n-1)[t^{n-1}]S^n$. The change-of-variables rule is valid for a formal substitution with nonzero linear coefficient; it can also be checked directly on Laurent monomials. In the analytic setting the same calculation follows from Cauchy’s integral formula on sufficiently small circles. Replacing t by $t_0 + w$, and S by ϕ_y , yields the coefficient formula of Theorem 7.1; a nonunit constant is allowed when one uses the same residue argument with $w = \delta\phi_y(w + t_0)$.

B Repository provenance and limits of the comparison

The repository was inspected on 29 September 2026. The relevant live path was `Analysis/Transseries`, not the older path under `Analysis/FabiusFunction` used in some earlier research notes. The retrieved Transseries subtree listing had Git tree identifier `81058854a185ab903d9012f6de6f93da345b31d6`. This identifies the inspected listing, not a claim that every file was read or that the whole repository was frozen at that state.

The principal content sources were the transseries group README and the README of the *Action Accumulation and Nonlinear Inversion* package [2, 3]. The former describes formal, asymptotic, and analytic interfaces and their qualifications. The latter provides the closest inspected comparison for hidden flat oscillations and nonlinear transport, including an explicit warning about the general Hahn-realization question. The canonical multi-volume sources and the new research packages were not audited theorem by theorem.

An indexed repository code search for “Stieltjes” returned relevant uses in other material; an indexed search for “Wigert” in the Transseries path returned no matches. Neither search proves absence of an equivalent theorem or historical novelty. The local `SOURCE_NOTES.md` records the inspected paths and the primary literature used. No repository file was changed, and no external article is bundled in this package.

References

- [1] V. Reshetnikov, *ProveIt*, repository, inspected 29 September 2026. [GitHub: Vladimir-Reshetnikov/ProveIt](#).
- [2] V. Reshetnikov, ProveIt contributors and research artifacts, *Series and transseries*, group README, inspected 29 September 2026. [GitHub: Series and transseries README](#).
- [3] ProveIt research artifact, *Action Accumulation and Nonlinear Inversion: A Laplace-measure calculus, hidden oscillations, and limits of Hardy-field realization*, package README and editorial amendments, 29 September 2026. [GitHub: Action Accumulation and Nonlinear Inversion README](#).
- [4] G. D. Lin and J. Stoyanov, *Moment determinacy of powers and products of non-negative random variables*, Journal of Theoretical Probability **28** (2015), 1337–1353. doi:10.1007/s10959-014-0546-z. Author manuscript: [arXiv:1403.0301](#).
- [5] T. S. Chihara, *A characterization and a class of distribution functions for the Stieltjes–Wigert polynomials*, Canadian Mathematical Bulletin **13** (1970), 529–532. doi:10.4153/CMB-1970-098-7.
- [6] J. S. Christiansen, *The moment problem associated with the Stieltjes–Wigert polynomials*, Journal of Mathematical Analysis and Applications **277** (2003), 218–245. doi:10.1016/S0022-247X(02)00534-6.
- [7] J. S. Christiansen and E. Koelink, *Self-adjoint difference operators and classical solutions to the Stieltjes–Wigert moment problem*, Journal of Approximation Theory **140** (2006), 1–26. doi:10.1016/j.jat.2005.11.010.
- [8] L. Di Vizio and C. Zhang, *On q -summation and confluence*, Annales de l’Institut Fourier **59** (2009), 347–392. doi:10.5802/aif.2433. [arXiv:0709.1610](#).
- [9] N. Nikolaev, *Gevrey asymptotic implicit function theorem*, L’Enseignement Mathématique **70** (2024), 251–282. doi:10.4171/LEM/1061.
- [10] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, Chapter 17, especially Section 17.8, Jacobi’s triple product. <https://dlmf.nist.gov/17.8>. Consulted 29 September 2026.
- [11] [Editorial addition, ProveIt, 2026-09-29.] *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and inversion*, the canonical volume of the series-and-transseries group: `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`.
- [12] [Editorial addition, ProveIt, 2026-09-29.] *Certified Inversion Through the q -to-1 Transition*, research package filed in batch 46: `Analysis/Transseries/docs/series-and-transseries/Certified_Inversion_q_to_1_Transition/q_transition.tex`.
- [13] [Editorial addition, ProveIt, 2026-09-29.] *A Spectral Representation for the Inverse Digamma Function*, research package filed in batch 46: `Analysis/Transseries/docs/series-and-transseries/Inverse_Digamma_Spectral_Representation/inverse_digamma_spectral.tex`.